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RESEARCH ARTICLE

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Ensemble deep learning framework for landslide susceptibility mapping and road vulnerability index development in the Chittagong Hill Tracts, Bangladesh

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ABSTRACT

Landslides pose a persistent threat to life, livelihoods, and critical infrastructure in the Chittagong Hill Tracts (CHT) of Bangladesh, a region marked by steep slopes, intense monsoonal rainfall, and increasing anthropogenic pressure. This study presents a comprehensive landslide susceptibility assessment for the monsoon-dominated Chittagong Hill Tracts (CHT), addressing data scarcity challenges. An ensemble model integrating Convolutional Neural Networks, Deep Neural Networks, and Long Short-Term Memory networks was developed using ten standardized 30-meter resolution geo-environmental variables. A balanced dataset of 5,082 samples was generated through spatially stratified random sampling. Spatial-block 10-fold cross-validation demonstrated strong model performance with an F1-score of 0.8923 and an AUC of 0.9558, outperforming individual models. Key drivers identified include annual rainfall and proximity to fault lines, highlighting the role of hydro-tectonic factors. Additionally, a Road Vulnerability Index based on six physical parameters revealed that 13% of the regional road network is highly vulnerable to landslides. These findings offer high-resolution, interpretable susceptibility and infrastructure risk maps to support disaster risk reduction strategies, contributing to Sustainable Development Goals 11 and 13. This framework advances understanding of landslide hazards and infrastructure vulnerability in data-limited mountainous regions.

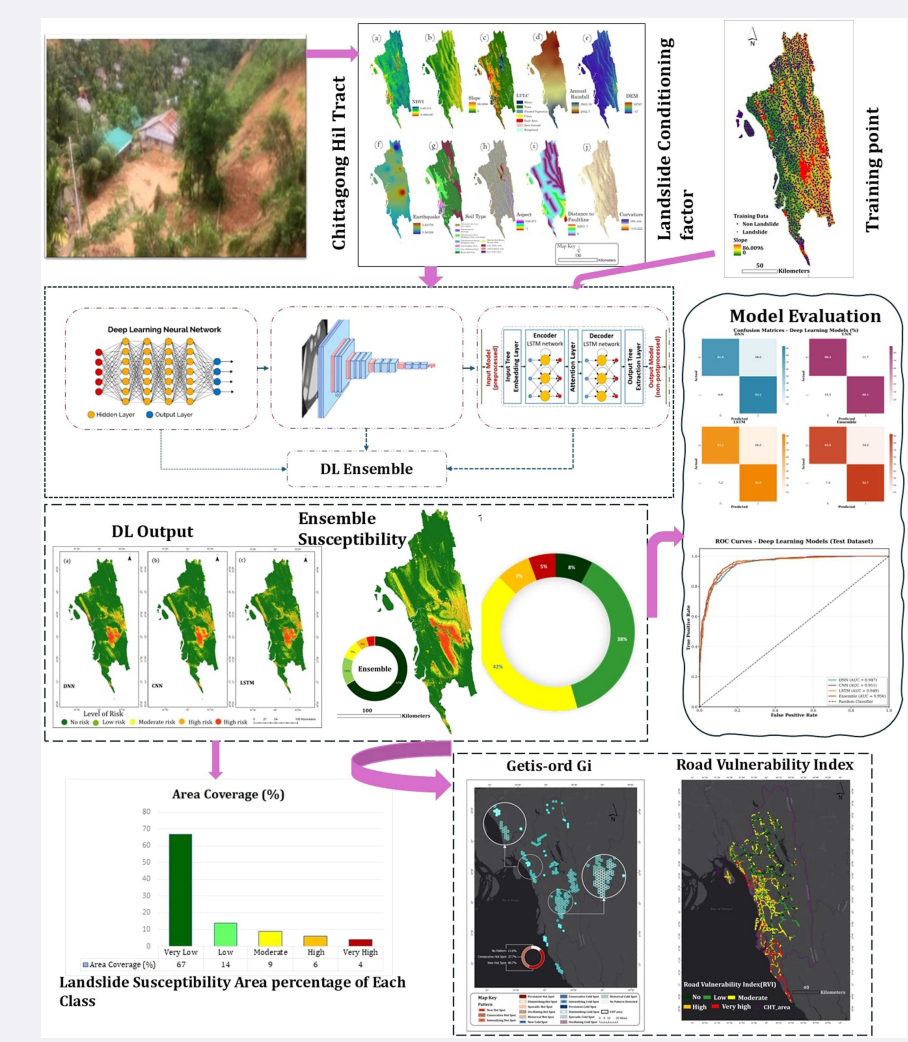
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Ensemble deep-learning framework for landslide susceptibility mapping and Road Vulnerability Index (RVI) development in the Chittagong Hill Tracts, Bangladesh (Figure).

1 Introduction

Landslides are one of the most pervasive and destructive natural hazards in mountainous and hilly terrains, causing significant loss of life, damage to critical infrastructure, and long-term ecological consequences (Islam et al. 2017; Abdo 2025). Globally, landslides are often triggered by intense rainfall, seismic events, and anthropogenic disturbances such as unplanned urbanisation, deforestation, and land-use changes (Osman and Barakbah 2006; Chisty 2014). In Bangladesh, the southeastern Chattogram Hill Tracts are especially vulnerable due to steep slopes, heavy monsoonal rainfall, and increasing human encroachment into unstable terrain (Ahmed 2015; Rabby and Li 2020b). Over the past two decades, this region has experienced repeated landslide events, with reports indicating that over 250 people have died since 2007 due to slope failures in Chattogram alone (Ali et al. 2018). Compounding the physical hazard is the socio-economic vulnerability of approximately half a million people who live in informal settlements on or near unstable slopes (Galli and Guzzetti 2007; Pardeshi et al. 2013; Rabby and Li 2020b). The vulnerability and risk associated with landslides arise from the interplay between physical susceptibility factors like steep slopes, loose soils, and triggering events and human exposure, particularly through poorly planned infrastructure such as road networks and hillside settlements (Galli and Guzzetti 2007; Bammou et al. 2025). In the CHT, rainfall stands out as the primary trigger, often intensified by seismic shaking, altered slope morphology, degraded soil characteristics, vegetation loss, and human modifications

including road cuts that disrupt drainage and weaken materials (Pardeshi et al. 2013; Chisty 2014; Gupta et al. 2016; Dai et al. 2024; Huang 2024). These elements transform natural processes into disasters, especially in areas where roads serve as essential links for development yet amplify hazards by creating unstable cut slopes.

While various landslide susceptibility models have been developed ranging from statistical techniques to artificial neural networks (Guzzetti et al. 2006), they often rely on dense historical landslide inventories and consistent geospatial datasets that are not always available or reliable in data-constrained environments like Bangladesh. Furthermore, popular frameworks such as the Analytical Hierarchy Process (AHP), though widely adopted, may not be effective in this context due to bias in expert judgement and the method's limitation in handling non-linear relationships among variables (Myronidis et al. 2016). To address these gaps, this study poses two key research questions: (1) How can an ensemble of deep learning models (CNN, DNN, LSTM) improve the accuracy and interpretability of landslide susceptibility mapping in data-constrained mountainous regions? (2) In what ways does a novel Road Vulnerability Index (RVI) enhance the assessment of infrastructure exposure by linking susceptibility to road-specific factors like terrain steepness and precipitation? In this study, we aim to bridge this methodological and contextual gap by developing a geospatially integrated landslide susceptibility model tailored to the Chattogram Hilly Area. The model leverages a comprehensive set of environmental, geological, and infrastructural variables including elevation, slope, curvature, NDVI, precipitation, proximity to rivers and roads, soil type, and earthquake occurrence standardised to a 30-metre resolution using authoritative sources such as the USGS, NASA, OpenStreetMap, and the Survey of Bangladesh.

A key novelty of this work is the introduction of the Road Vulnerability Index (RVI), a composite indicator that quantifies the susceptibility of road segments to landslide impact by considering proximity to slope failure zones, local terrain steepness, precipitation intensity, and road alignment. The RVI draws on physical principles of slope stability, incorporating factors such as annual rainfall (a primary trigger via soil saturation), distance to fault lines (for tectonic weakening), and land use/land cover (influencing vegetation reinforcement), supported by studies on hazard-infrastructure interactions (van Westen et al. 2008; Habumugisha et al. 2022).

This tailored metric addresses a critical knowledge gap by directly linking road infrastructure vulnerability with landslide risk in a region where transportation corridors are vital yet highly exposed to natural hazards. Unlike existing models that treat roads as passive indicators, our RVI allows for proactive identification of high-risk segments and supports strategic planning for slope stabilisation and route realignment. The objectives of this paper are twofold: (1) to produce a robust landslide susceptibility map for the Chittagong Hill Tracts by first evaluating the individual performance of three deep learning models Convolutional Neural Networks (CNN), DeepNeural Networks (DNN), and Long Short-Term Memory (LSTM) networks and then combining their predictions through a weighted ensemble approach that maximises accuracy in a data-limited setting; and (2) to introduce and apply a new Road Vulnerability Index (RVI) that translates susceptibility outputs into practical risk rankings for road segments, enabling authorities to pinpoint the most exposed corridors for urgent stabilisation and redesign. Although deep learning ensembles are increasingly used in landslide studies, the specific ensemble developed here is novel: it is the first to integrate a CNN (operating on 9×9 pixel patches), a DNN (capturing global non-linear interactions), and an LSTM (treating monthly rainfall as a 12-month time series at each pixel) within a single weighted-average framework, an architecture that, to our knowledge, has not previously been applied to landslide susceptibility mapping in monsoon-dominated, tectonically active, data-scarce mountain regions. These contributions deliver not only higher predictive reliability than standalone models but also a ready-to-use tool for disaster managers, offering a straightforward, replicable workflow that can be directly adopted in other landslide-prone, data-scarce mountain regions across South and Southeast Asia.

1.1 Overview of the methodological workflow

The landslide susceptibility modelling and Road Vulnerability Index (RVI) development follow an integrated deep learning framework specifically designed for the data-constrained environment of the Chittagong Hill Tracts (CHT). The workflow comprises six main stages (see Figure 1): (i) acquisition and harmonisation of multi-source geospatial data, (ii) rigorous preprocessing and spatially stratified sampling

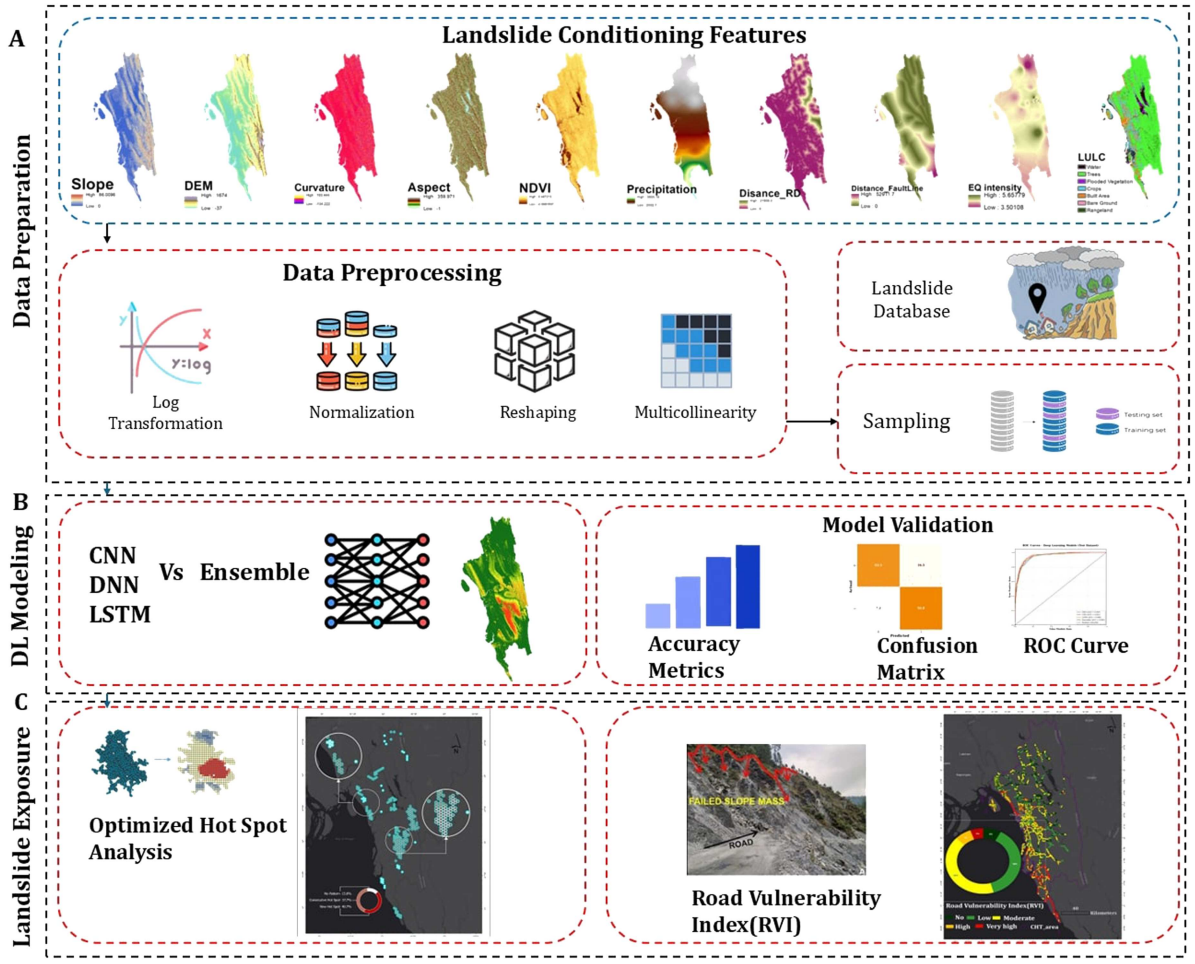


Figure 1. Methodology flowchart. This Section show step by steps system to calculate Landslide Susceptibility and RVI.

to handle skewness, scale differences, and spatial bias, (iii) individual training and evaluation of three deep learning architectures Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Long Short-Term Memory (LSTM) networks, (iv) construction of a weighted ensemble model that combines the strengths of the individual learners, (v) model validation using spatial block 10-fold cross-validation and multiple performance metrics including Kappa and Matthews correlation coefficient, and (vi) derivation of the Road Vulnerability Index (RVI) followed by exposure analysis of the road network. All raster layers were standardised to a 30 m resolution in WGS84 UTM Zone 45 N. Detailed descriptions of data sources and quality control are provided in Section 2.2 and Table 1; preprocessing and sampling strategies in Sections 2.3–2.4.1; deep learning architectures and ensemble weighting in Sections 2.5–2.6; validation protocols in Section 2.7; SHAP interpretability in Section 2.8; and RVI calculation, theoretical justification, and sensitivity analysis in Section 2.9.

1.2 Landslide conditioning features (LCFs) preparation

For landslide susceptibility modelling, eleven conditioning features (LCFs) were selected based on a comprehensive review of existing literature, regional geological characteristics, and data availability for the Chattogram Hilly Area. These factors are elevation, slope, aspect, plan and profile curvature, distance to rivers, distance to roads, distance to fault lines, annual rainfall, NDVI (annual maximum), soil type, and land use/land cover (LULC). Earthquake epicentres were processed separately as kernel density. All factors were standardised to 30 m resolution to ensure spatial consistency.

Table 1. Landslide conditioning factors (LCFs), sources, references, temporal coverage, resolution after processing, and quality/validation notes.

No.	Factor	Source	Temporal coverage	Original resolution	Resolution after processing	Quality/validation notes	Reference
1	Elevation	USGS EarthExplorer + Survey of Bangladesh	2000–2010	30 m	30 m	Vertical RMSE = 4.8 m (120 ground control points)	Rabby and Li (2020a)
2	Slope, Aspect	Derived from DEM	–	–	30 m	Slope in degrees; aspect in degrees from north	Rabby and Li (2019); Bammou et al. (2025); Rabby and Li (2020a)
3	Plan & Profile Curvature	Derived from DEM	–	–	30 m	Unitless	Rabby and Li (2020b)
4	Distance to rivers	OpenStreetMap + Survey of Bangladesh	2024	Vector	30 m (Euclidean)	Buffer accuracy checked against high-resolution imagery	Mourin et al. (2018)
5	Distance to roads	OpenStreetMap	2025	Vector	30 m (Euclidean)	Major roads verified with field photos (latest OSM export January 2025)	van Westen et al. (2008); Chisty (2014)
6	Distance to fault lines	Geological Survey of Bangladesh	2020	Vector	30 m (Euclidean)	Digitised from 1:50 000 geological maps	Huang (2024)
7	Annual rainfall	NASA TRMM 3B43 v7 + CHIRPS	TRMM: 1998–2019; CHIRPS: 1981–2024	0.25°	30 m (IDW, power = 2)	Cross-validated against 12 local rain-gauge stations ($R^2 = 0.91$, MAE = 187 mm)	Islam et al. (2017)
8	NDVI (annual max)	Landsat 8-9 OLI (Google Earth Engine)	2013–2024	30 m	30 m (native)	Cloud-free annual maximum value composite; QA band used for masking (full 2024 season included)	Rabby and Li (2020b)
9	Soil type	Bangladesh Agricultural Research Council (BARC) Maps Portal	2020	Vector (polygon)	30 m (nearest neighbour)	Reclassified into 11 drainage/texture classes directly from BARC national soil map	Mourin et al. (2018)
10	Land use/ land cover	ESA WorldCover 2021	2021	10 m	30 m (majority)	Overall accuracy **74.4%** (ESA validation report)	Ganerød et al. (2023)
11	Earthquake epicentres	USGS earthquake catalogue	1900–2024	Point	30 m (kernel density)	Magnitude ≥ 4.0 within 200 km radius; bandwidth 10 km (catalogue updated December 2024)	Chowdhury (2024); Huang (2024)

All raster datasets were resampled to a common 30 m grid using bilinear interpolation for continuous variables and majority/nearest-neighbour for categorical variables, then projected to WGS84 UTM Zone 45 N. Initial cleaning, annual NDVI compositing using Landsat 8–9 surface reflectance (Collection 2, 2013–2024), and rainfall gap-filling were performed in ArcGIS Pro. Soil type data, obtained from the Bangladesh Agricultural Research Council (BARC) Maps Portal, was converted from vector to raster in ArcGIS Pro 3.5. Final alignment, quality cheques, and Euclidean distance calculations were completed in ArcGIS Pro 3.5. Rainfall interpolation used Inverse Distance Weighting (IDW, power = 2) with cross-validation against local meteorological stations ($R^2 = 0.91$). The spatial distribution of all LCFs is presented in Figure 2.

1.3 LCFs Preprocessing

To prepare the Landslide Conditioning Features (LCFs) for robust application across a range of predictive modelling approaches, including machine learning, deep learning, and ensemble methods, a comprehensive preprocessing pipeline was implemented. This process included log transformation, normalisation, data reshaping, and multicollinearity analysis. Each step was designed to enhance model performance by addressing distributional biases, improving feature comparability, and aligning the data structure with the requirements of advanced algorithms. Log transformation was applied to variables prone to extreme values and skewed distributions, such as elevation and rainfall frequency. This step reduces the influence of

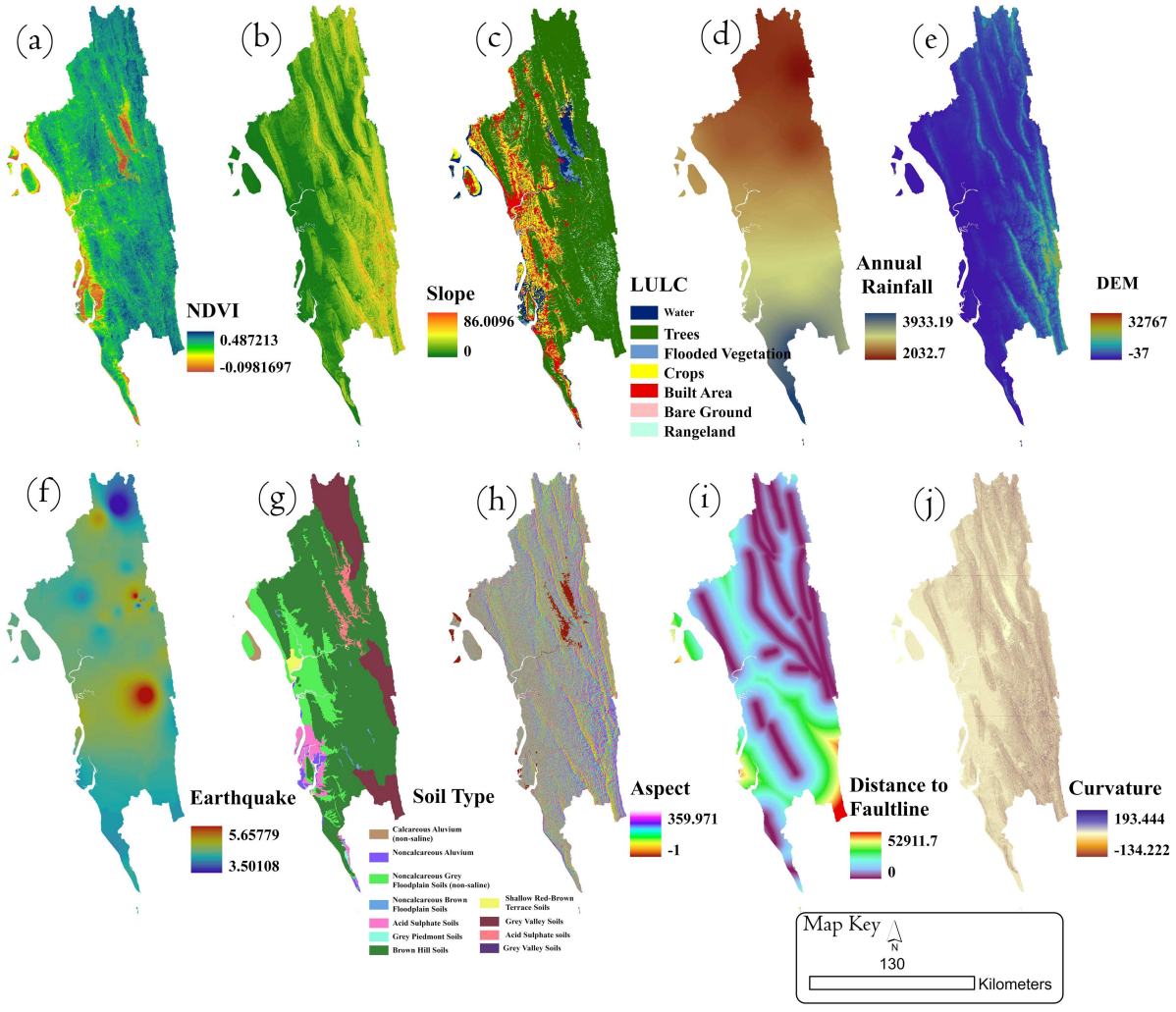


Figure 2. Landslide Conditioning Features (LCFs) including (a) NDVI, (b) Slope, (c) LULC, (d) Annual Rainfall, (e) DEM, (f) Earthquake, (g) Soil Type, (h) Aspect, (i) Distance to Faultline, (j) Curvature.

outliers and improves the symmetry of feature distributions, which is particularly beneficial for models sensitive to input variance. The transformation followed the standard form:

$$\log Transformed = \log(\delta + \gamma)$$

where δ is the original feature value and γ is a small constant used to avoid undefined values due to zero or negative inputs (Roy et al. 2019).

After log transformation, min–max normalisation was applied to scale all features between 0 and 1. This uniform scaling is essential for gradient-based learning algorithms and ensemble techniques, ensuring that no single feature dominates due to differences in units or magnitude (Cui et al. 2025). Two normalisation strategies were used, depending on the direction of the feature's influence on landslide susceptibility. For directly correlated features (e.g. rainfall frequency), positive normalisation was applied:

$$Positive\ Normalization = \frac{\delta - \min(\delta)}{\max(\delta) - \min(\delta)}$$

$$Negative\ Normalization = \frac{\max(\delta) - \delta}{\max(\delta) - \min(\delta)}$$

This dual approach ensured that the scaled values preserved the correct directionality of each feature's contribution to landslide risk.

The normalised features were then reshaped to match the input format required by different modelling techniques (Roy et al. 2019). Deep learning models especially Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks benefit from structured 2D input that retains spatial or sequential relationships in the data. Reshaping was therefore applied as needed, using NumPy, and the processed arrays were stored in compressed. npz format to improve data handling efficiency during training and inference. For the Convolutional Neural Network (CNN), the data were reshaped into 9×9-pixel patches to preserve spatial context. For the Long Short-Term Memory (LSTM) network, rainfall values were arranged as time-series sequences of the past 12 months at each pixel location. By maintaining a uniform data structure and reducing redundancy, the preprocessing pipeline ensured that ensemble models could effectively integrate predictions from heterogeneous learners. Finally, a multicollinearity analysis was performed to detect and manage highly correlated input variables. Features with strong linear correlations (Pearson's $|r| > 0.8$) were either excluded or consolidated to prevent redundancy and reduce the risk of overfitting. This step was particularly important for both individual learners and ensemble frameworks, as multicollinearity can distort model interpretation and degrade generalisation performance.

1.3.1 Hyperparameter selection

All landslide conditioning factors were treated as numeric. Missing values were imputed using the mean strategy, and features were z-score normalised based on the training data in each cross-validation fold to avoid information leakage. Hyperparameters for the DNN, CNN, and LSTM models were optimised via grid search combined with stratified 10-fold cross-validation Described in Table 2. Early stopping and learning rate reduction on plateau were applied during training to prevent overfitting. The best hyperparameter configuration for each model was selected based on the highest validation ROC-AUC, with accuracy reported secondarily. All stochastic processes, including weight initialisation and data shuffling, used a fixed random seed (random_state = 42) to ensure reproducibility. Model training was conducted on Google Colab Pro + using an Intel Xeon 2.20 GHz CPU, NVIDIA Tesla T4 GPU (16 GB), and 52 GB RAM.

1.4 Multicollinearity between LCFs

Multicollinearity refers to the occurrence when predictor variables (in this case, Landslide Conditioning Features (LCFs) are highly correlated with each other, which can cause bias in the final model. High correlation among input features can lead to redundancy, affecting the model's performance and interpretability (Dormann 2013). When two or more features exhibit strong linear relationships, they provide similar information to the model, potentially leading to unstable parameter estimates, inflated standard errors, and reduced predictive accuracy (O'Brien 2007).In this study, an initial set of 11 LCFs was

Table 2. Hyperparameter specifications for the deep learning models.

Model	Hyperparameter	Search range/Options	Selected value
CNN	Learning rate	0.0001–0.01	0.001
	Batch size	32, 64, 128	32
	Number of conv layers	1–2	2
	Filters per layer	32–64	64–32
	Dropout	0.2–0.5	0.3
	Optimiser	Adam, RMSprop	Adam
DNN	Number of dense layers	3–5	5
	Neurons per layer	512-256-128-64-32	512-256-128-64-32
	Dropout	0.2–0.5	0.3
	Learning rate	0.0001–0.01	0.001
	Optimiser	Adam	Adam
LSTM	Number of LSTM layers	1–2	2
	Units per layer	50–200	64–32
	Dropout	0.2–0.5	0.3
	Learning rate	0.0001–0.01	0.001
All models	Epochs	100–200	150 (early stopping)
	Early stopping patience	10–20	20
	Loss function	Binary cross-entropy	Binary cross-entropy
	Validation split	20%	20%

prepared, comprising: elevation (DEM), slope, aspect, curvature, distance to fault lines (Distance F), distance to rivers, distance to roads, soil type, land use/land cover (LULC), annual rainfall (Annual Rai), and maximum rainfall (rmax). To assess multicollinearity, a comprehensive correlation analysis was conducted using Pearson's correlation coefficient. Variables exhibiting correlation coefficients with absolute values exceeding 0.70 were considered to indicate strong multicollinearity, as this threshold is widely recognised in the literature for identifying problematic collinearity that may compromise model performance (Midi et al. 2010; Dormann 2013).

The initial correlation analysis revealed that the maximum rainfall (rmax) feature exhibited an exceptionally high correlation (r = 0.94) with annual rainfall (Annual Rai), as illustrated in Figure 3a using a circular chord diagram. This visualisation technique, adapted from genomic data representation

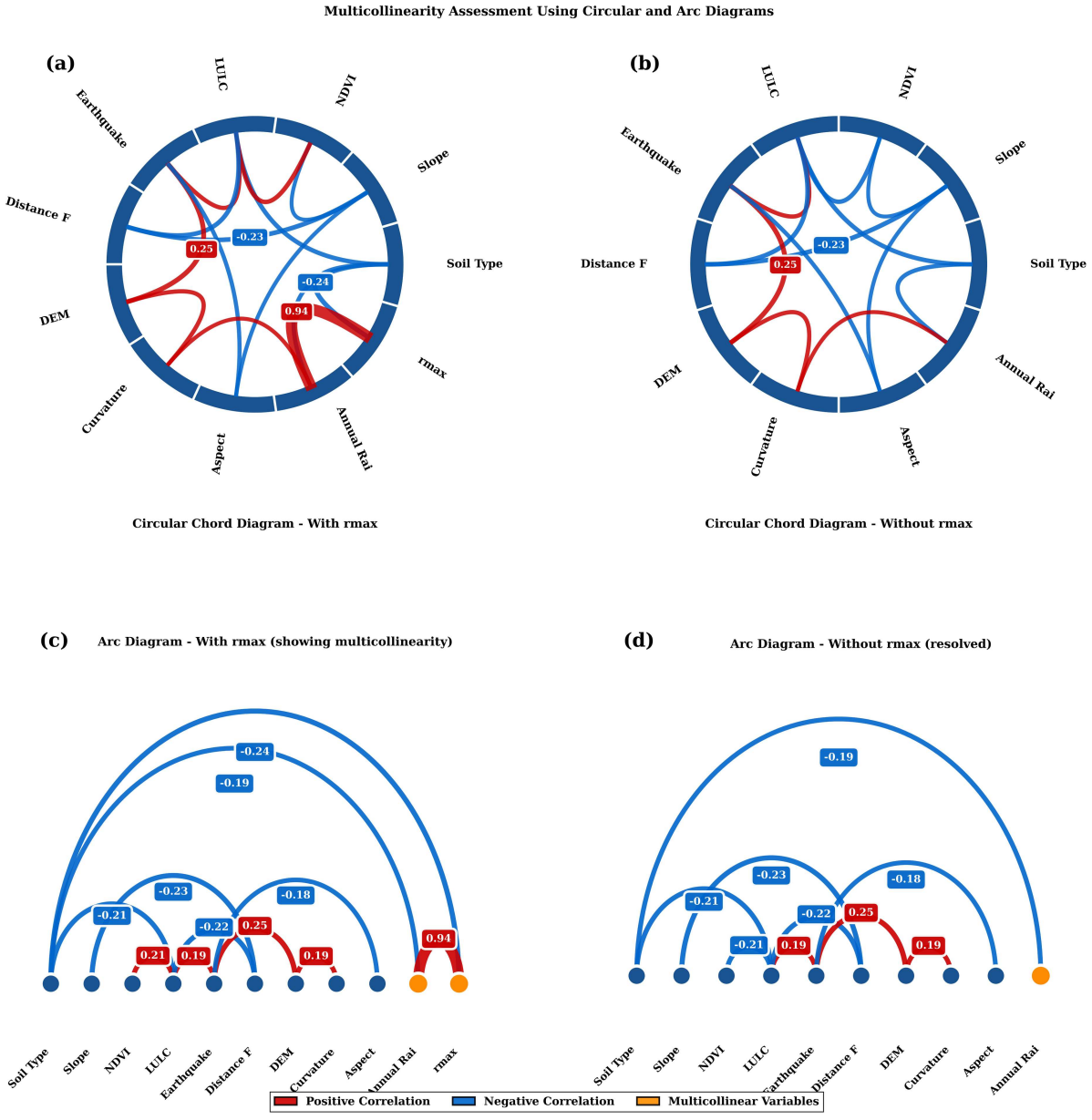


Figure 3. Multicollinearity assessment of landslide conditioning factors (a) Circular chord diagram of the initial 11 features showing high multicollinearity between Annual Rai and rmax (r = 0.94); (b) Chord diagram after removing rmax, demonstrating resolved multicollinearity with all correlations below the 0.70 threshold; (c) Arc diagram of the initial feature set highlighting the problematic correlation (orange nodes indicate multicollinear variables); (d) Arc diagram of the final 10 features showing improved independence.

(Krzywinski 2009), provides an intuitive depiction of the correlation structure by positioning variables along a circular perimeter with curved ribbons connecting correlated pairs. The thickness and colour intensity of the ribbons represent the strength and direction of correlations, respectively, with red indicating positive and blue indicating negative relationships. The prominent thick red ribbon between Annual Rai and rmax clearly demonstrates their near-perfect linear relationship, which could distort the model's ability to independently assess the contribution of each feature.

To resolve this multicollinearity issue, the rmax variable was systematically removed from the dataset, as it provided redundant information already captured by annual rainfall. Following this removal, a re-evaluation of the remaining 10 features was conducted to ensure no residual multicollinearity remained. The updated correlation matrix (Figure 3b) confirmed that no variable pair exceeded the correlation threshold of 0.70, with the strongest remaining correlation being between earthquake frequency and DEM ($r = 0.25$). This demonstrates that all remaining features are sufficiently independent for inclusion in the modelling framework. To further validate the independence of the selected features, an arc diagram representation (Figures 3c and 3d) was employed to visualise the correlation network structure. In this diagram, variables are arranged linearly, with arcs connecting correlated feature pairs. The height and thickness of each arc represent the correlation strength, while orange nodes highlight variables identified in the multicollinearity assessment (Annual Rai and rmax). Figure 3c clearly shows the dense arc connecting the orange nodes in the initial dataset, while Figure 3d demonstrates the resolved correlation structure after rmax removal. These complementary visualisation approaches provide multiple perspectives on the correlation structure, enhancing the transparency and interpretability of the feature selection process. The final set of 10 independent LCFs comprising soil type, slope, NDVI, LULC, earthquake frequency, distance to fault lines, DEM, curvature, aspect, and annual rainfall was retained for training the three deep learning models (CNN, DNN, LSTM) and the weighted ensemble. This rigorous multicollinearity assessment ensures that each feature contributes unique and non-redundant information to the landslide susceptibility modelling framework, thereby enhancing model stability, interpretability, and predictive performance.

1.4.1 Sampling strategy and bias mitigation

A total of 5,082 sample points (2,541 landslide and 2,541 non-landslide) were selected for analysis (see in Figure 4). The column 'Value' was used as the landslide indicator. To avoid spatial clustering and ensure even geographic coverage, a spatially stratified random sampling approach was applied using a regular 5-km grid across the study area. Within each grid cell, landslide and non-landslide points were randomly chosen. Table 3 outlines The Global Moran's I value of 0.844 ($p < 0.01$) indicated significant positive spatial autocorrelation, confirming that landslide events tend to cluster in space. To mitigate bias in sample selection, the stratified sampling ensured that both landslide and non-landslide points were evenly distributed across the study area. The representativeness of the non-landslide samples was evaluated using a Chi-square goodness-of-fit test ($\chi^2 = 1.00$, $p = 0.64$), which confirmed that these points adequately captured the full range of environmental conditions, including slope, elevation, and land-use/land-cover (LULC) types. Since natural landslide pixels account for less than 3% of the total area, a balanced 1:1 class ratio was adopted to prevent the model from being biased toward the majority class.

1.5 Deep learning-based landslide susceptibility modelling

In regions with limited data availability, such as the Chittagong Division, Bangladesh, the absence of a comprehensive disaster management database has historically posed challenges for effective landslide risk assessment. To address this, the first step in this study was the creation of a landslide hazard database, which compiled historical landslide events and related geo-environmental data. This database will support both the current study and future landslide risk management efforts. The dataset used for modelling includes a range of Landslide Conditioning Features (LCFs), such as topographical factors (elevation, slope), environmental factors (NDVI, rainfall) (Habumugisha et al. 2022). These LCFs were then used to train various deep learning models to classify areas based on their susceptibility to landslides. The deep learning algorithms employed in this study include Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Long Short-Term Memory (LSTM) networks. These algorithms were selected due to their strong performance in handling large, high-dimensional datasets (Abdo 2025; Wu et al. 2025).

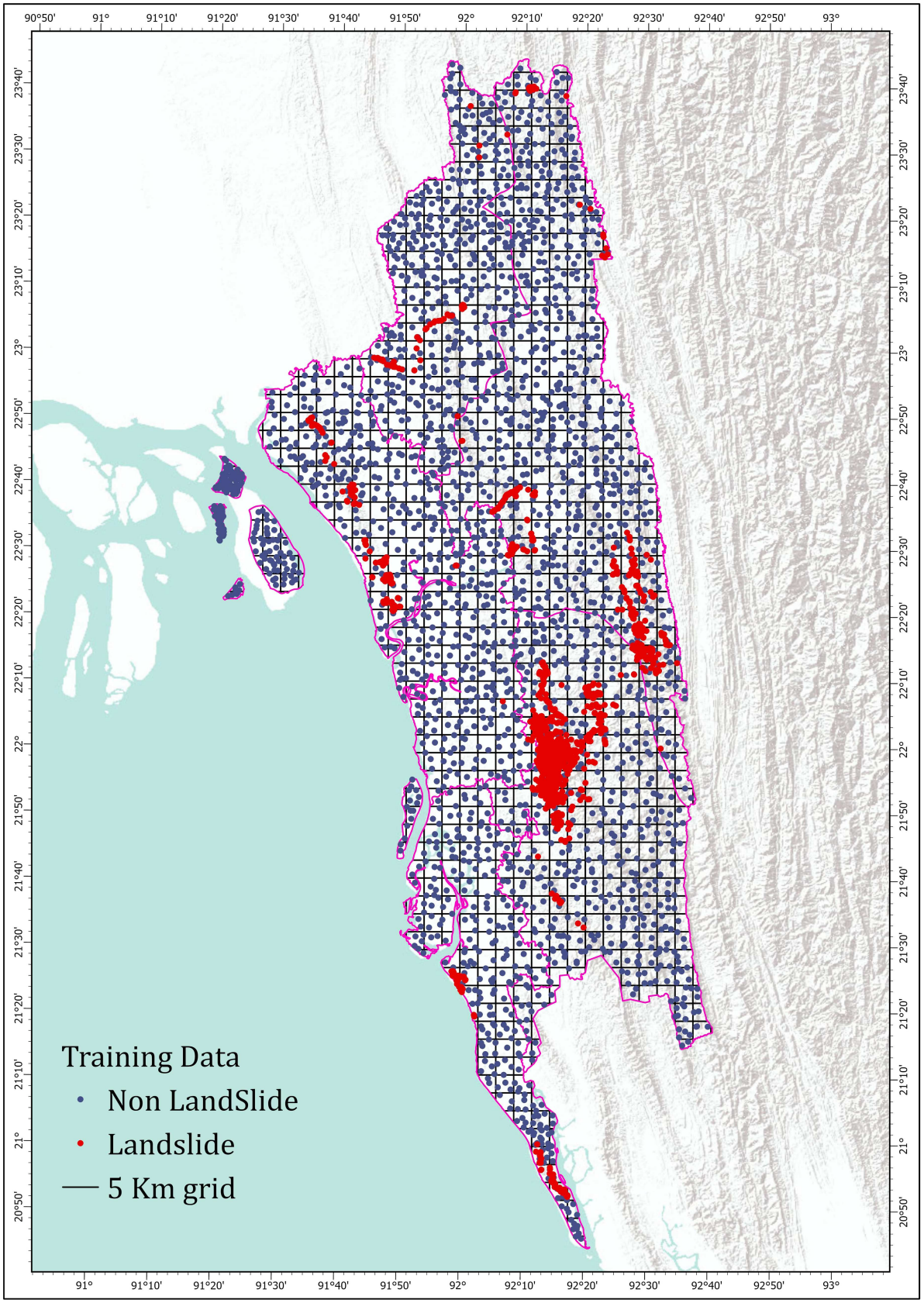


Figure 4. Spatial distribution of the 5,082 stratified random sample points (2,541 landslide in red, 2,541 non-landslide in blue) overlaid on the 5-km stratification grid.

Table 3. Sampling strategy and bias-mitigation statistics.

Item	Details
Total sample points	5,082
Landslide points	2,541
Non-landslide points	2,541
Sampling method	Spatially stratified random sampling on 5-km grid
Global Moran's I	0.844 ($p < 0.01$)
Chi-square test (class proportionality)	$p = 0.64$
Natural landslide pixel proportion	<3%
Final class ratio	1:1 (balanced)

CNNs are particularly effective at capturing spatial relationships in geospatial data, DNNs excel at modelling complex nonlinear relationships among features, and LSTMs are well-suited for capturing sequential dependencies in time-series data, such as rainfall patterns. Additionally, an ensemble model was created by combining the outputs of these individual deep learning models to improve overall prediction accuracy. The model validation process involved using performance metrics such as accuracy, precision, recall, F1-score, and Area Under the Curve (AUC) to evaluate the models' ability to generalise. The deep learning models, including the ensemble, showed excellent predictive performance with robust results. The final output of the study includes landslide susceptibility maps generated by these deep learning models, providing probabilistic estimates of landslide risk at a pixel level. These outputs can be used for assessing landslide hazard and informing future research in landslide risk assessment (Ganaie et al. 2022).

1.6 Ensembling of deep learning models using weighted average

To enhance landslide susceptibility prediction, an ensemble approach was implemented by combining multiple deep learning models Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Long Short-Term Memory networks (LSTM) (Fang et al. 2020; Biswas et al. 2025). Each model produces a probability $P_i(x)$ of landslide occurrence at a given location x. The final ensemble prediction, $\hat{P}_{ensemble}(x)$, is calculated as a weighted average of these probabilities, where the weight ω_i for each model reflects its performance. The ensemble prediction is given by:

$$\hat{P}_{ensemble}(x) = \sum_{i=1}^{N} \omega_i \quad P_i(x)$$

The weights ω_i are calculated based on each model's error rate $error_i$ as follows:

$$\omega_i \quad = \frac{1 - error_i}{\sum_{j=1}^{3} (1 - error_i)}$$

This approach assigns higher weights to models with lower error rates, ensuring they contribute more significantly to the final prediction. By leveraging the complementary strengths of CNN, DNN, and LSTM, the ensemble produces a more robust and accurate landslide susceptibility map. This method reduces the bias and limitations of individual models, improving overall prediction reliability.

1.7 Validation of deep learning and ensemble models

Validating deep learning models is essential to ensure their reliability and accuracy in predicting landslide susceptibility. Relying on a single accuracy metric can be misleading, as it may fail to capture issues like overfitting, underfitting, or imbalanced data. To provide a more complete evaluation, this study used multiple performance metrics to assess the models. We focused on deep learning approaches including Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), Long Short-Term Memory networks (LSTM), and an ensemble model combining these individual models' predictions(Fang et al. 2020; Lv et al. 2022). A key challenge in geospatial datasets is class imbalance, where landslide-prone and non-landslide areas are not equally represented. To prevent spatial autocorrelation bias, we applied spatial block 10-fold cross-validation using 5 km × 5 km blocks. Each block was excluded once as the test set,

while the remaining blocks formed the training set. This method reduces overfitting and promotes better generalisation by training and testing the models on varied data splits. In addition to accuracy, precision, recall, F1-score, and AUC, we also report Cohen's Kappa and Matthews Correlation Coefficient (MCC) to better handle class imbalance Detail description are mentioned in Table 4. All supporting diagnostic plots are available in Supplementary Figs. S1–S13.

1.8 Feature importance

Deep learning models such as Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), Long Short-Term Memory networks (LSTM), and their ensemble versions are powerful tools for complex prediction tasks but are often regarded as 'black boxes' because it is difficult to interpret how they make decisions (Lundberg and Lee 2017). To overcome this challenge and gain insights into the role of different landslide conditioning factors (LCFs) in the model's predictions, we employed Shapley Additive explanations (SHAP). SHAP assigns an importance score to each feature for every prediction, allowing us to fairly and clearly understand how much each factor contributes to the output without bias toward any single feature. (Sun et al. 2024). SHAP values were calculated on the final weighted ensemble predictions using the spatially stratified sample described in Section 2.4.1. To explore feature influence, we used SHAP summary plots and waterfall plots. The summary plots provide an overall picture of which features most strongly impact landslide susceptibility predictions, while others such as vegetation cover (NDVI) might reduce the risk. The force plots offer detailed explanations for individual predictions by illustrating how each feature pushes the model's decision toward or away from classifying an area as landslide-prone. Through these SHAP analyses applied to our deep learning models and their ensemble, we identified the most influential features contributing to landslide risk, improving model transparency and interpretability.

1.9 Road vulnerability index (RVI) calculation methods

The Road Vulnerability Index (RVI) was calculated by integrating multiple environmental factors that influence landslide susceptibility. Six key factors annual rainfall, distance to fault lines, land use/land cover (LULC), soil type, slope aspect, and curvature were selected based on their demonstrated relevance to slope stability and data availability.

Each factor R_i was first reclassified into discrete vulnerability classes. To ensure comparability and avoid bias due to differing units and ranges, min-max normalisation was applied:

$$N_i = \frac{R_i - R_i^{\min}}{R_i^{\max} - R_i^{\min}}$$

where N_i is the normalised score of factors i, $R_i^{\min}$, $R_i^{\max}$ and are the minimum and maximum values of the reclassified factor i across the study area. This scales each factor to a continuous range [0, 1]. To

Table 4. Details of accuracy metrics used.

Metric	Equation	Description
Accuracy	$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$	The proportion of correct predictions (True Positives + True Negatives) over the total predictions.
F1-Score	$F1\ \ Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$	The harmonic means of Precision and Recall, balancing the trade-off between them.
Precision	$Precision = \frac{TP}{TP+FP}$	The ability of the model to correctly identify positive instances (True Positives).
Recall	$Recall = \frac{TP}{TP+FN}$	The ability of the model to correctly identify all relevant positive instances.
ROC Curve	ROC Curve: TPR vs. FPR	A plot of the True Positive Rate (TPR) against the False Positive Rate (FPR) at various thresholds.
AUC	$AUC = \int_0^1 ROC\ Curve$	The area under the ROC curve, quantifying the model's ability to distinguish between positive and negative classes.
Kappa	$Kappa = \frac{P_0 - P_e}{1 - P_e}$	Measures agreement corrected for chance; values close to 1 indicate excellent agreement.
MCC	$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP(TN+FN)}}$	Ranged from −1 to +1; +1 perfect prediction, 0 random, −1 inverse prediction.

• TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.
• N = Total number of accuracy metrics used, = individual accuracy metrics.

compute the composite RVI for each road segment, a weighted average of the normalised factors was calculated:

$$RVI = \Sigma(\omega_i \times N_i)$$

where ω_i is the assigned weight of factor i, reflecting its relative importance derived from permutation feature importance in Deep learning models.

The weights are normalised such that:

$$\Sigma\omega_i = 1$$

The weighted average RVI values thus range between 0 (lowest vulnerability) and 1 (highest vulnerability).

For interpretation and risk categorisation, RVI values were classified into five vulnerability classes based on the following thresholds:

$$\text{Class}\begin{cases}\text{No Vulnerability}, & \text{If RVI} \le 0.20\\ \text{Low Vulnerability}, & \text{if } 0.20 < \text{RVI} \le 0.40\\ \text{Moderate Vulnerability}, & \text{if } 0.40 < \text{RVI} \le 0.60\\ \text{High Vulnerability}, & \text{if } 0.60 < \text{RVI} \le 0.80\\ \text{Very High Vulnerability}, & \text{if RVI} > 0.80\end{cases}$$

The proportion of roads falling into each class was calculated as:

$$P_c = \left(\frac{N_c}{N_{total}}\right) \times 100$$

Where P_c is the percentage of road segments in class c, N_c is the count of roads in class c, and N_{total} is the total number of road segments analysed.

All computations and classifications were automated using ArcPy within ArcGIS Pro 3.5, ensuring reproducibility and consistency. This weighted average approach allows integration of heterogeneous environmental factors into a single, normalised index that effectively characterises landslide vulnerability along the road network.

1.9.1 Theoretical justification of RVI factors

The six factors included in the Road Vulnerability Index are supported by both physical principles of slope stability and the results of the ensemble model. Annual rainfall is the primary triggering factor because heavy monsoon rainfall raises pore-water pressure and reduces effective stress in soil, thereby lowering shear strength (Iverson 2005). Proximity to fault lines increases susceptibility because faults produce fractured and weakened rock mass that fails more easily during rainfall or seismic shaking (Huang 2024). Land use/land cover affects root reinforcement and surface runoff; deforestation and bare soil remove vegetation roots and increase runoff velocity (Sidle and Ochiai 2013). Soil type controls drainage capacity; clay-rich soils retain water longer and lose strength rapidly when saturated (van Westen et al. 2008). Slope aspect influences soil moisture regime through differential solar radiation; south-facing slopes dry faster in the dry season, form cracks, and then fill with water during monsoon (van Westen et al. 2008). Curvature determines whether water converges (concave areas) or diverges (convex areas) along the road alignment, affecting local saturation and erosion (Chisty 2014). These six factors dominate the permutation feature importance ranking of the final ensemble model (Table 8), with annual rainfall, distance to fault lines, and land use/land cover occupying the top three positions. Weights used in the RVI were obtained by normalising the permutation importance values of exactly these six factors, ensuring full consistency between the susceptibility model and the road-specific vulnerability index.

2 Results

Model training and performance evaluation were conducted for landslide susceptibility prediction across the Chittagong Hill Tracts (CHT), Bangladesh, utilising Google Colab Pro + . The implementation was carried out in Python, employing scikit-learn for data preprocessing, PyTorch and TensorFlow with Keras for deep learning architectures, and ArcGIS Pro for generating and rendering the final susceptibility maps. As detailed in Tables 5 and 6, the ensemble model consistently outperformed the individual deep learning models CNN, DNN, and LSTM in both classification accuracy and generalisation capability, underscoring the enhanced reliability achieved through model integration.

2.1 Landslide susceptibility modelling and accuracy assessment

Model validation was conducted using 10-fold spatial block cross-validation with 5 km × 5 km grid blocks to minimise spatial autocorrelation bias and ensure robust generalisation across the topographically complex Chittagong Hill Tracts. Table 5 reports the mean and standard deviation of accuracy, precision, and recall across the 10 folds. Table 6 and Figure 5 present the performance metrics obtained when each model was trained on the full dataset and evaluated on an independent test set. Table 7 lists the per-fold accuracy values for all models. The ensemble model outperformed all individual models across all metrics. On the final test set (Table 6), it achieved an accuracy of 0.894, precision of 0.896, recall of 0.894, F1-score of 0.894, and AUC-ROC of 0.956. It also recorded the highest Kappa coefficient (0.789) and Matthews correlation coefficient (0.790). Among the standalone models, LSTM achieved the highest accuracy (0.890) and AUC-ROC (0.951), followed by CNN (accuracy = 0.881, AUC = 0.947) and DNN (accuracy = 0.869, AUC = 0.945). During cross-validation (Table 3), the ensemble recorded a mean accuracy of 0.893 ± 0.012, mean precision of 0.894 ± 0.014, and mean recall of 0.893 ± 0.014, with the lowest standard deviation in all three metrics. Detailed model diagnostics are provided in Supplementary Figs. S1–S6.

The confusion matrices (Figure 6) show that the ensemble model correctly classified the highest number of landslide and non-landslide pixels. Training accuracy was 0.934, and test accuracy was 0.892. The receiver operating characteristic (ROC) curves (Figure 7) indicate that the ensemble model achieved the highest area under the curve (AUC = 0.956). The ensemble ROC curve lies above the curves of all individual models across the full range of false positive rates.

2.2 Landslide susceptibility disparities in Southeast Bangladesh

The landslide susceptibility maps (LSMs) produced by the three deep learning models DNN, CNN, and LSTM are presented in Figure 8. Each model was developed using the same set of ten landslide conditioning factors and applied across the Chittagong Hill Tracts (CHT) region. Figure 8a shows the output of the Deep Neural Network (DNN), Figure 8b displays the results from the Convolutional Neural Network (CNN), and Figure 8c illustrates the susceptibility patterns derived from the Long Short-Term Memory (LSTM) network. To make the results more interpretable, the continuous susceptibility values generated by each model were classified into five distinct categories: very low, low, moderate, high, and

Table 5. Mean and standard deviation of performance metrics across 10-fold spatial block cross-validation.

Model	Mean accuracy	± Std	Mean precision	± Std	Mean recall	± Std
DNN	0.867	0.016	0.851	0.024	0.892	0.029
CNN	0.860	0.027	0.860	0.027	0.861	0.054
LSTM	0.878	0.020	0.848	0.029	0.924	0.014
Ensemble	0.893	0.012	0.894	0.014	0.893	0.014

Table 6. Final test set performance of deep learning and ensemble models.

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC	Kappa	MCC	Specificity	Sensitivity
DNN	0.869	0.871	0.869	0.869	0.945	0.738	0.739	0.835	0.903
CNN	0.881	0.882	0.881	0.881	0.947	0.763	0.763	0.858	0.904
LSTM	0.890	0.892	0.890	0.890	0.951	0.780	0.782	0.855	0.925
Ensemble	0.894	0.896	0.894	0.894	0.956	0.789	0.790	0.866	0.923

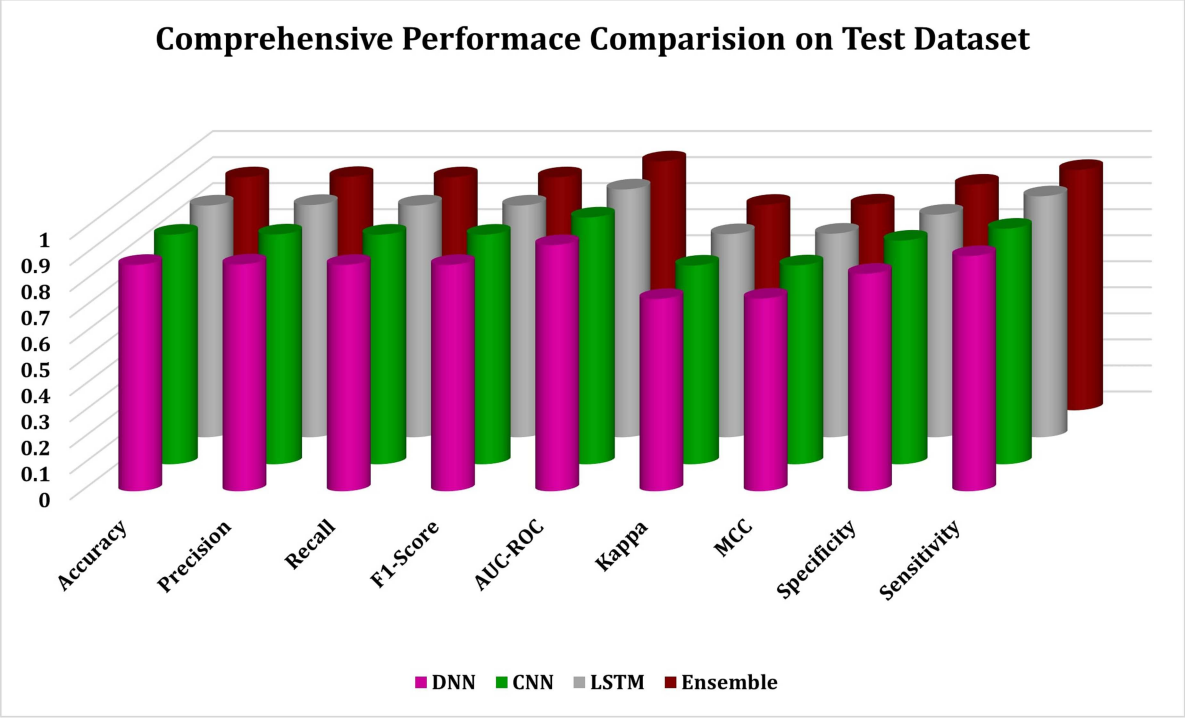


Figure 5. Accuracy metrics of all deep learning and ensembling methods.

Table 7. Per-fold accuracy during 10-fold spatial block cross-validation.

Fold	DNN	CNN	LSTM	Ensemble
1	0.882	0.865	0.848	0.896
2	0.879	0.865	0.865	0.894
3	0.840	0.789	0.843	0.875
4	0.885	0.905	0.899	0.912
5	0.862	0.857	0.899	0.894
6	0.862	0.874	0.868	0.889
7	0.874	0.865	0.899	0.897
8	0.842	0.848	0.887	0.887
9	0.856	0.865	0.885	0.890
10	0.887	0.868	0.887	0.896

very high. This classification was based on quantile breaks, where each class represents 20 percent of the value distribution. In other words, the 'very low' class includes the lowest 20 percent of values, followed by 'low' (21–40%), 'moderate' (41–60%), 'high' (61–80%), and 'very high' (81–100%). While this method does not reflect the actual physical area under each class, it provides a consistent and statistically balanced way to compare outputs across different models. Although the overall spatial trends are broadly similar, each model captures the susceptibility patterns with slight variations. High-risk zones are consistently observed in the steep and fragmented hill slopes of southeastern Bandarban and parts of Rangamati, while low and very low susceptibility areas are more common in the relatively gentle terrains of Khagrachari and its surroundings. The differences in model outputs reflect how each neural network architecture interprets spatial features and relationships. These results highlight the importance of evaluating multiple models to understand the range of susceptibility predictions, which lays the foundation for selecting or combining models for more accurate hazard assessments.

2.3 Ensemble-based landslide susceptibility mapping

To generate a more reliable and spatially consistent landslide susceptibility map, we combined the outputs of the three deep learning models DNN, CNN, and LSTM usingthe weighted average ensemble approach

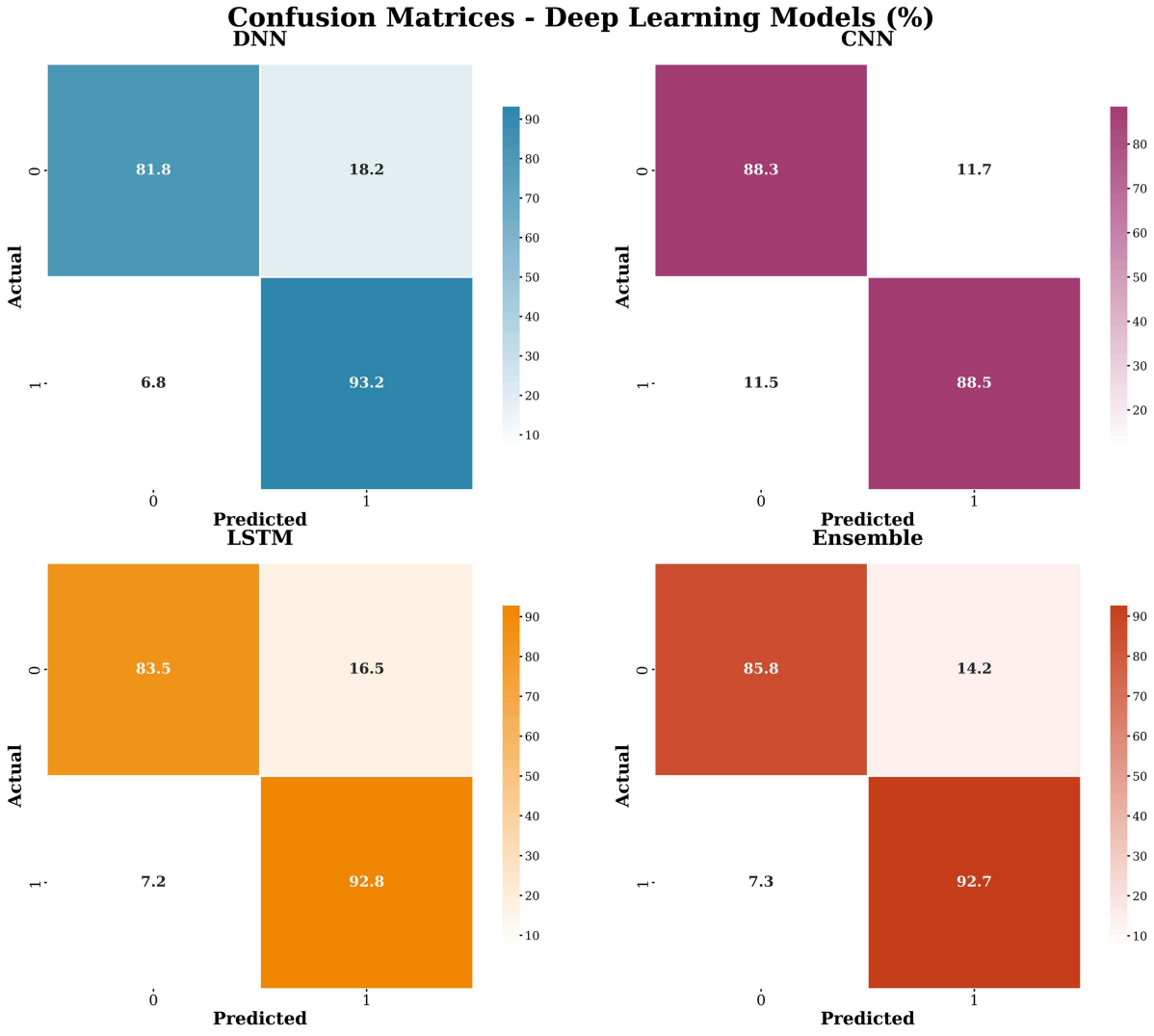


Figure 6. Confusion matrix of ensemble model.

described in Section 2.6. This method blends the strengths of each model and gives higher weight to the better-performing ones. The ensemble-based susceptibility map, shown in Figure 9, We classified it into five categories using the same quantile-based method as the individual models: very low, low, moderate, high, and very high. This makes comparison easy and helps understand the risk pattern across the Chittagong Hill Tracts (CHT). Compared to the individual models, the ensemble map shows a smoother and more consistent. High-risk zones remain concentrated in the southeastern parts of Bandarban and southern Rangamati, where steep slopes and unstable terrain dominate. Low-risk areas are mainly in the flatter northwestern parts of Khagrachari. A closer look at the classified ensemble map reveals that about 67% of the region falls under the *very low* susceptibility category, while *low* and moderate classes account for roughly 14% and 9%, respectively. The remaining 6% and 4% of the area are classified as high and very high susceptibility zones. The figures show in Figure 9 percentage clearly. Even though most of the CHT is low risk, the small high-risk areas near roads and steep slopes need urgent attention and protection. Threshold optimisation results are shown in Supplementary Figs. S10–S13.

2.4 Influence of conditioning factors on landslide susceptibility

To explore the influence of individual conditioning factors on landslide susceptibility and enhance the interpretability of the ensemble model, feature importance analysis was carried out using SHAP (Shapley Additive Explanations). Figure 10 displays the mean absolute SHAP values for each factor, indicating overall

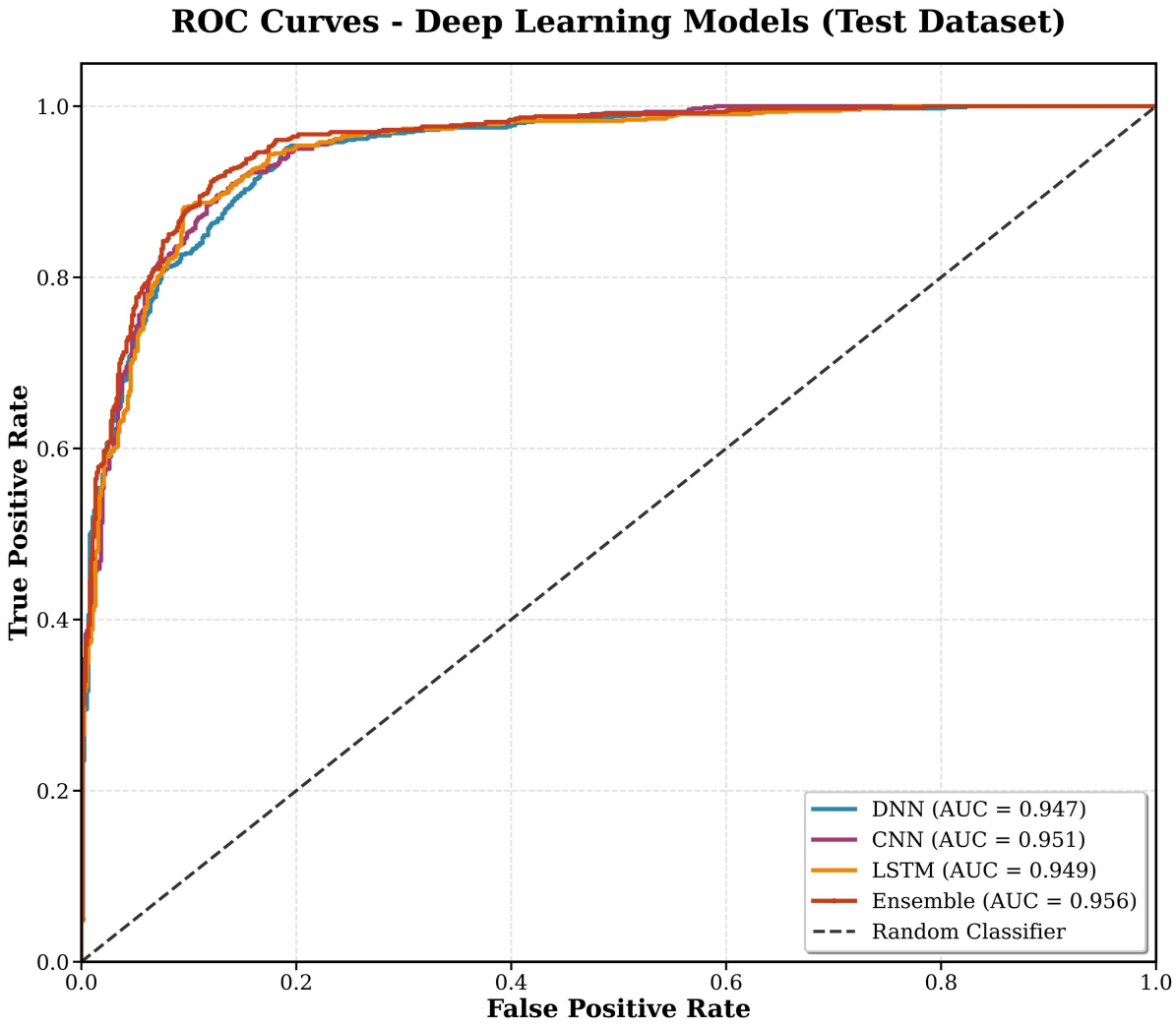


Figure 7. Receiver operating characteristics (ROC) Curve.

contribution to model output, with red indicating positive and blue negative impact directions. Figure 11 presents a waterfall explanation for a single high-risk prediction, showing how each factor incrementally adjusts the base model output (0.500) toward the final probability (1.136), with Annual Rainfall contributing the largest positive shift (+0.2622). Additional SHAP analyses, including dependence plots and local explanations for representative pixels, are provided in the Supplementary Materials (Figures S1–S5). SHAP importance rankings for the individual CNN, DNN, and LSTM models are also included in the Supplementary Materials (Tables S1–S3). The permutation importance analysis revealed that Annual Rainfall, Distance to Fault Lines, and Land Use/Land Cover (LULC) were the top three influencing factors (Table 8). Annual Rainfall recorded the highest impact (0.2622), followed by Distance to Fault Lines (0.2224) and LULC (0.0736). These results confirm that rainfall-induced saturation, tectonic proximity, and land surface modification are the dominant triggers in the study area. In contrast, factors like Slope, NDVI, Curvature, and Aspect showed very low importance (≤0.0129), indicating limited direct contribution to the ensemble predictions. However, SHAP force plots showed that slope and NDVI can have both positive and negative effects depending on location for example, steep slopes increase risk mainly under high rainfall and low vegetation, while dense NDVI reduces risk near fault zones. Permutation feature importance is reported in Supplementary Figs. S7–S9.

2.5 Hotspot analysis of landslide-prone zones in the Chittagong Hill Tracts

The spatial-temporal hotspot analysis reveals critical patterns of landslide vulnerability across the Chittagong Hill Tracts (CHT). Figure 12 shows the full picture.40.7% are new hot spots risk has

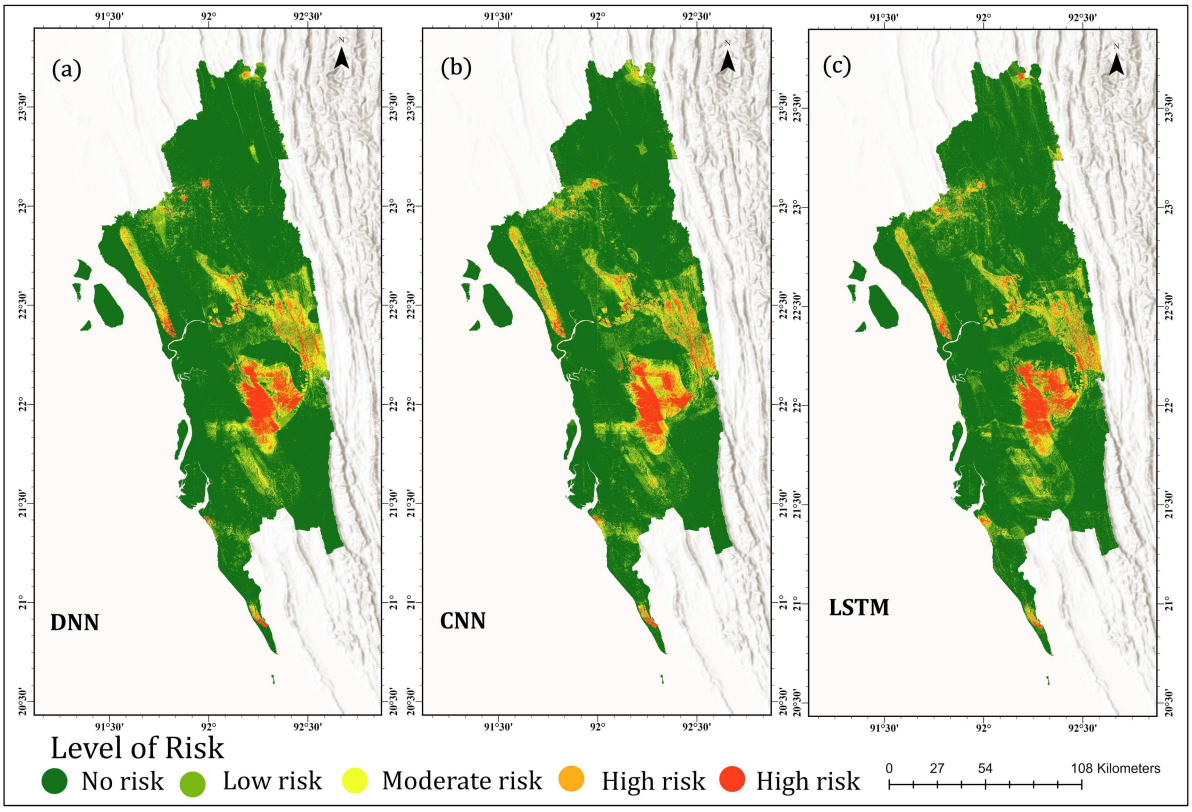


Figure 8. Landslide susceptibility maps for (a) DNN, (b) CNN, (c) LSTM and level of risk.

increased recently. 37.7% are consecutive hot spots high risk every year. Together more than 78% of the area has serious problems. These persistent and emerging clusters are especially concentrated in the southeastern parts of the CHT, including Bandarban, Khagrachari, and parts of Rangamati, where rugged terrain, deforestation, and expanding human settlement intersect. Smaller clusters are also visible in Cox's Bazar and adjacent coastal regions, suggesting a wider geographic spread of hazard-prone areas. Only 13.6% of the area showed no clear spatial pattern, pointing to relatively isolated events. This map proves landslide danger is growing fast in southeastern Bangladesh. We need urgent action –better land management, slope protection, and community safety plans in the worst zones.

2.6 Landslide vulnerability assessment of road networks in the Chittagong Hill Tracts

The landslide vulnerability assessment of the road network in the Chittagong Hill Tracts reveals a heterogeneous risk profile across critical transport corridors (see Figure 13). Out of the total road segments analysed, 484 roads (7.7%) were classified as having no significant vulnerability, while 2,370 roads (37.8%) exhibited low vulnerability, indicating relatively stable terrain in these areas. The majority of roads, totalling 2,618 segments (41.8%), fell into the moderate vulnerability category, reflecting widespread exposure to potential landslide hazards. Notably, 446 roads (7.1%) were identified as highly vulnerable, and 350 roads (5.6%) were classified as very highly vulnerable the most critical roads are Amir Hat–Md. Takir Hat Road, Bangladesh Sea Road, and Teknaf–Shamlapur Road. These are vital lifelines for local people and trade. The presence of these major roads in high-risk zones highlights the urgent need for focused mitigation efforts, including slope stabilisation and improved drainage infrastructure, to enhance safety and maintain connectivity. The donut chart in Figure 13 makes the risk levels easy to understand and helps plan urgent actions. This shows that more than 13% of roads are in high or very high danger zones and need immediate protection.

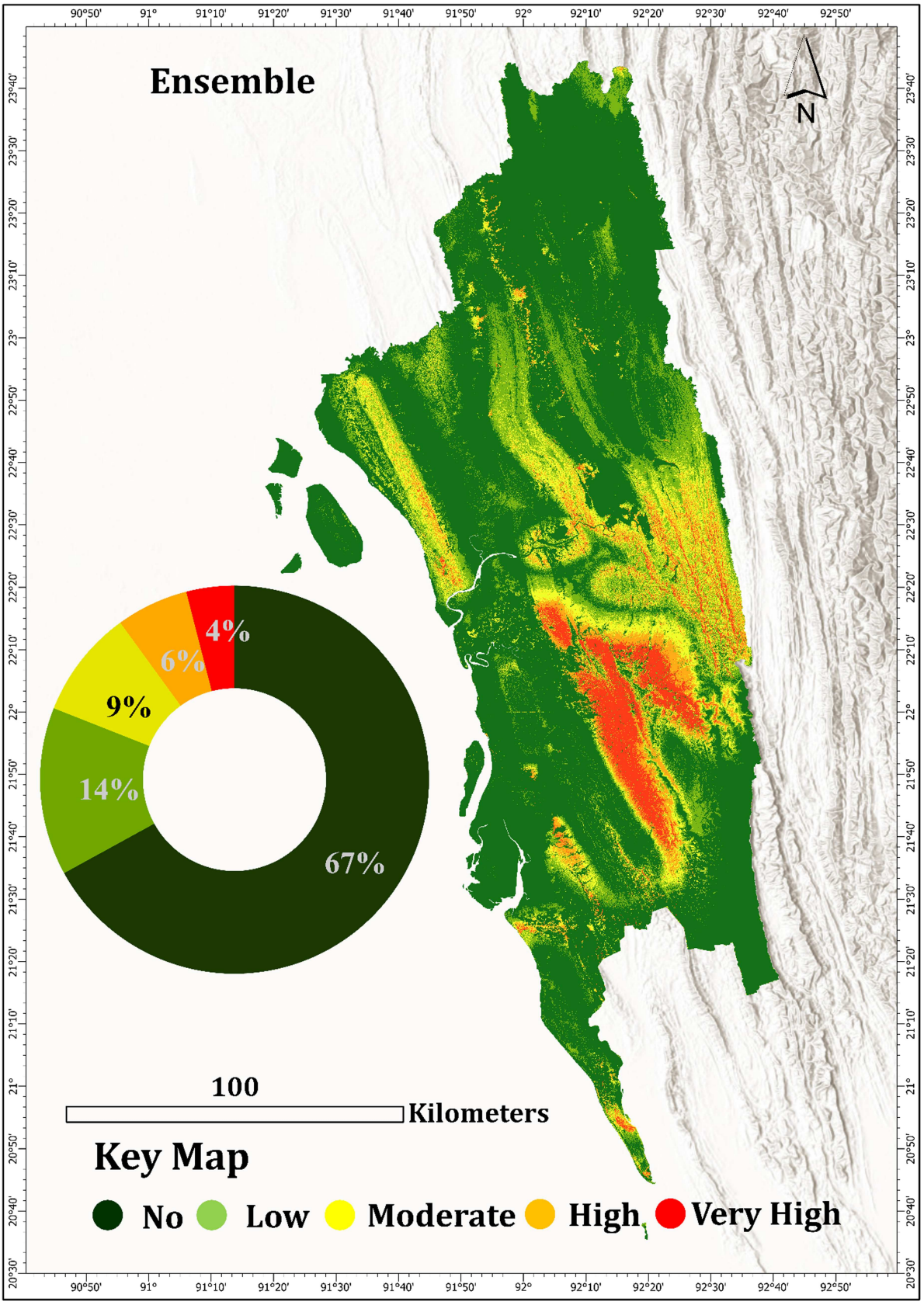


Figure 9. Landslide susceptibility maps for Ensemble model and the donut charts represent the distribution of landslide susceptibility classes across the entire study area.

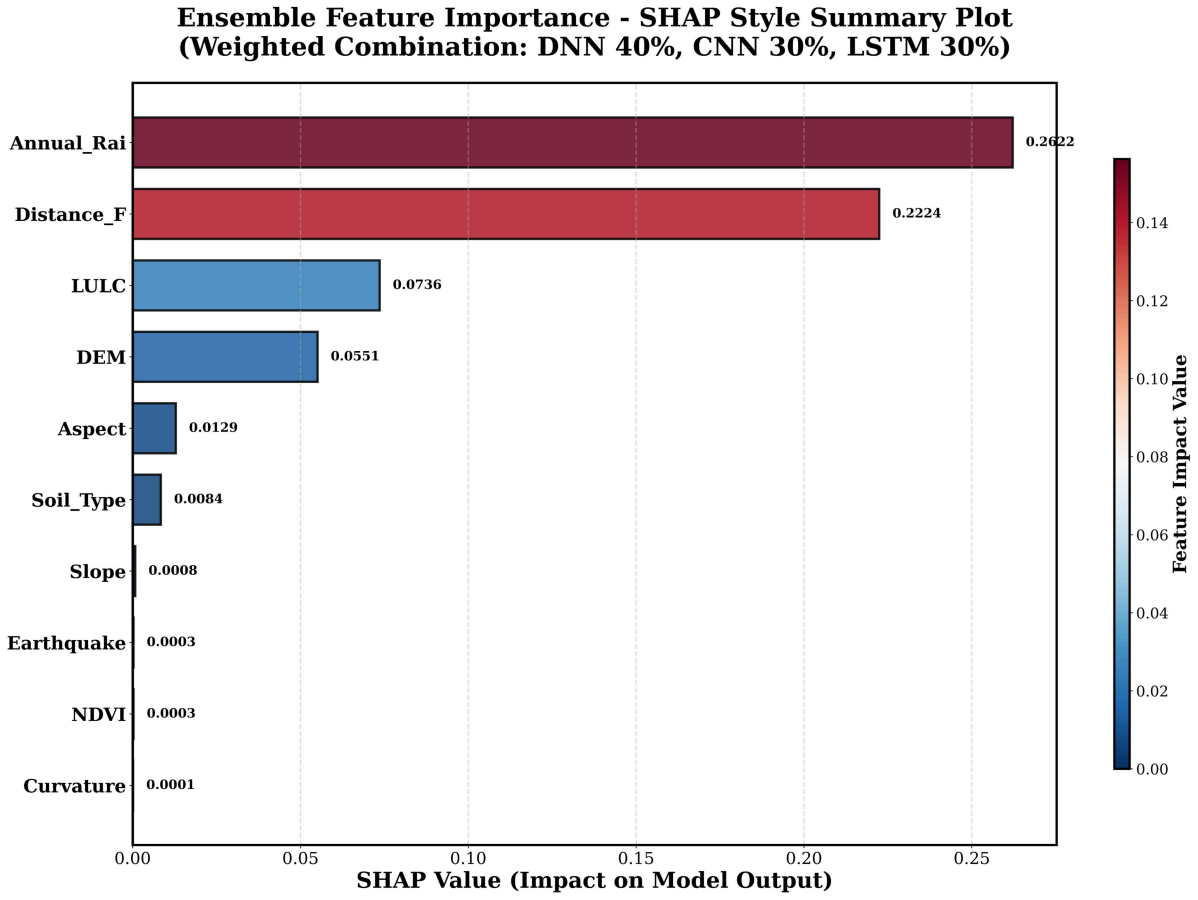


Figure 10. Feature importance of ensemble model and feature contribution summary plot.

3 Discussion

This study offers a significant advancement in landslide susceptibility modelling by applying a deep learning ensemble framework that is specifically adapted to the unique environmental and infrastructural context of the Chittagong Hill Tracts (CHT). The integration of Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Long Short-Term Memory (LSTM) networks into a unified ensemble model enabled superior predictive performance across key metrics, with the ensemble achieving an F1-score of 0.8923 and an AUC of 0. 9558. (Guzzetti et al. 2006). The applicability of this model to the CHT is further substantiated by its contextual accuracy. The CHT is characterised by steep, erosion-prone slopes, tectonic instability, and high monsoonal precipitation, making it acutely vulnerable to landslides (Islam et al. 2017). The SHAP analysis in this study demonstrated that Annual Rainfall, Distance from Fault Lines, and Land Use/Land Cover (LULC) were the most influential features contributing to landslide susceptibility predictions. These findings align with both historical disaster records and geological literature. For instance, the 2017 landslides in Bandarban and Rangamati, which claimed over 150 lives following more than 500 mm of rainfall in a single day, illustrate the pivotal role of intense precipitation in triggering slope failures (Secretariat et al. 2023). The high importance of rainfall identified by the model is thus a validation of its real-world relevance. In a similar vein, the importance of proximity to fault lines confirms the tectonic sensitivity of the region. The CHT sits along the Indo-Burman plate boundary, and geological studies have shown that this zone is under continuous seismic stress, which exacerbates slope instability, particularly during or after rainfall events (Huang 2024; Sun et al. 2024). That our model assigns substantial weight to fault line proximity through SHAP values underscores its ability to internalise subtle but critical geophysical risk factors. LULC, particularly the degradation of forest covers due to shifting cultivation and unregulated development, was also found to significantly affect susceptibility

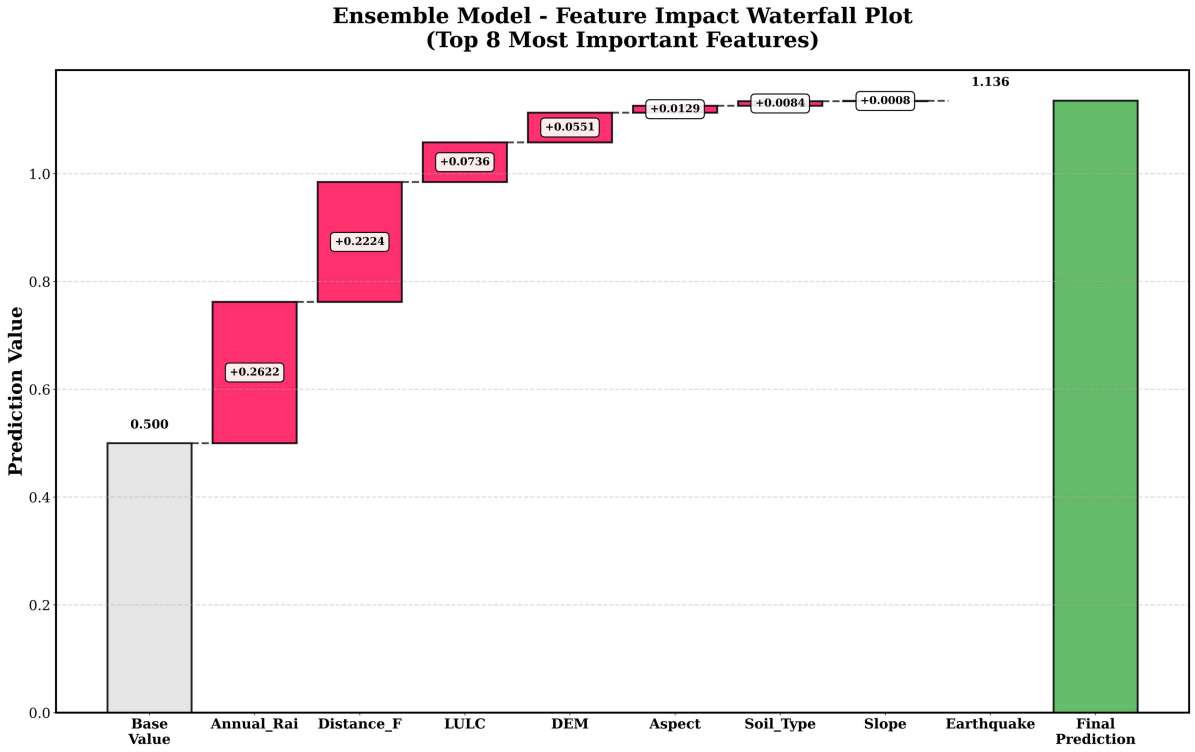


Figure 11. Ensemble model waterfall plot.

Table 8. Permutation-based absolute feature importance and ranking for the ensemble model.

Feature	Abs_importance	Rank
Annual_Rainfall	0.262163934	1
Distance_to_Fault	0.222426230	2
LULC	0.073573770	3
DEM	0.055081967	4
Aspect	0.012852459	5
Soil_Type	0.008393443	6
Slope	0.000786885	7
NDVI	0.000262295	8
Earthquake	0.000262295	9
Curvature	0.000131148	10

outcomes. According to (Ganerød et al. 2023), the CHT has lost over 11% of its tree cover since 2001, which directly diminishes slope stability by weakening root structures and increasing surface runoff. The model's ability to reflect this anthropogenic dimension of risk adds further credibility to its predictive and explanatory power. Our ensemble AUC of 0.9558 is higher than recent deep learning studies in monsoon-dominated mountains, such as 0.94 in the Western Ghats (Prakash 2024) and 0.93 in the Nepal Himalaya.

Surprisingly, conventional topographic variables such as slope, curvature, and aspect exhibited limited influence on the model's output. While these features are widely regarded as critical determinants in classical landslide modelling, their relative insignificance in this study likely stems from the ensemble model's capacity to capture complex, multi-factorial interactions where topographic features alone are insufficient predictors. (Ju et al. 2024). In our case, SHAP force plots revealed that the slope feature had both positive and negative influences depending on context, suggesting that it functions more as a conditional modifier than a standalone trigger in the CHT setting. This agrees with a global review of 100 + deep learning landslide studies that found slope importance drops when rainfall and land-use data are included (Merghadi 2020).

Another major contribution of this research is the development and application of a Road Vulnerability Index (RVI), which links landslide susceptibility predictions to the resilience of road infrastructure. In a

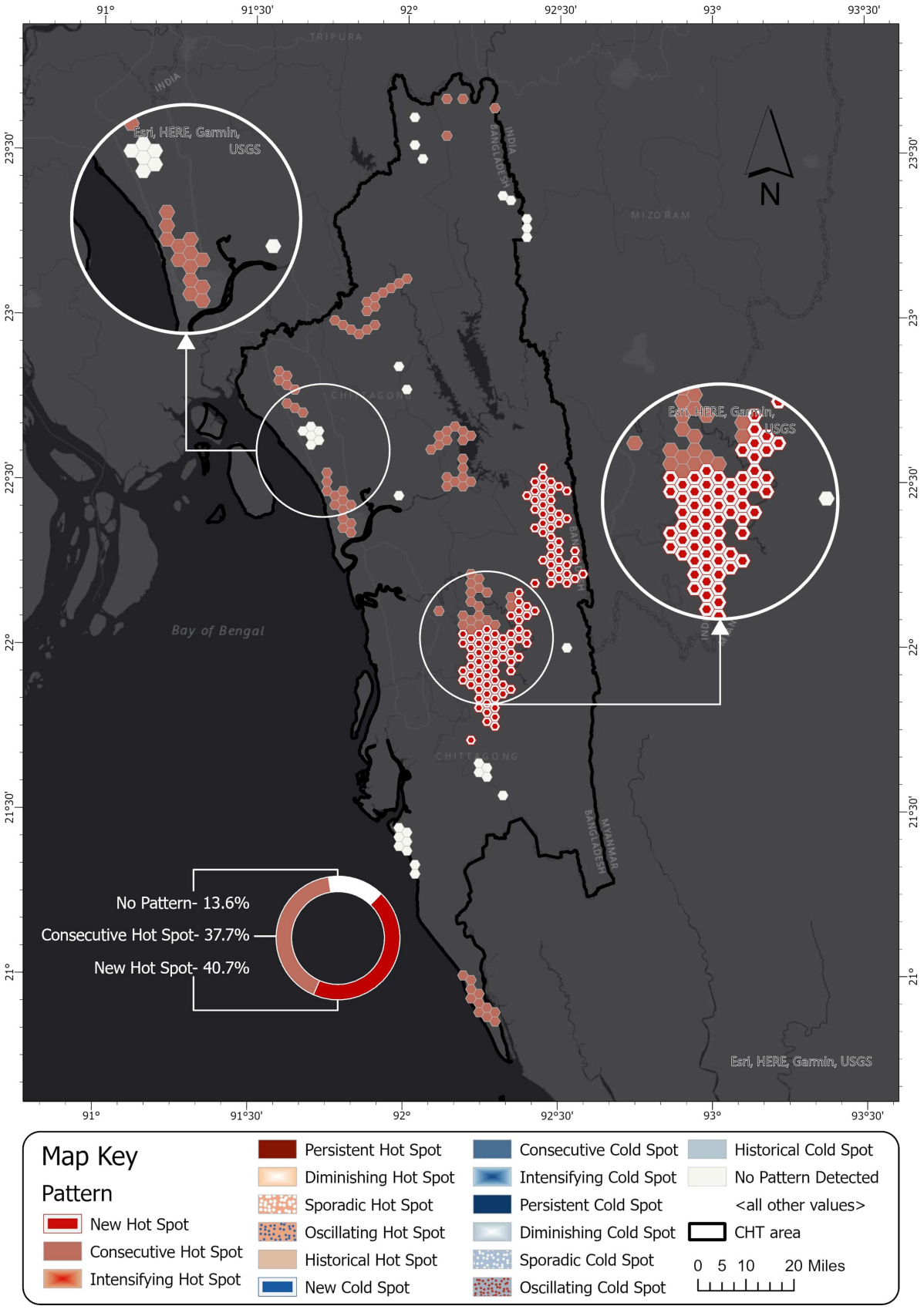


Figure 12. Emerging hotspot analysis for very high landslide classes.

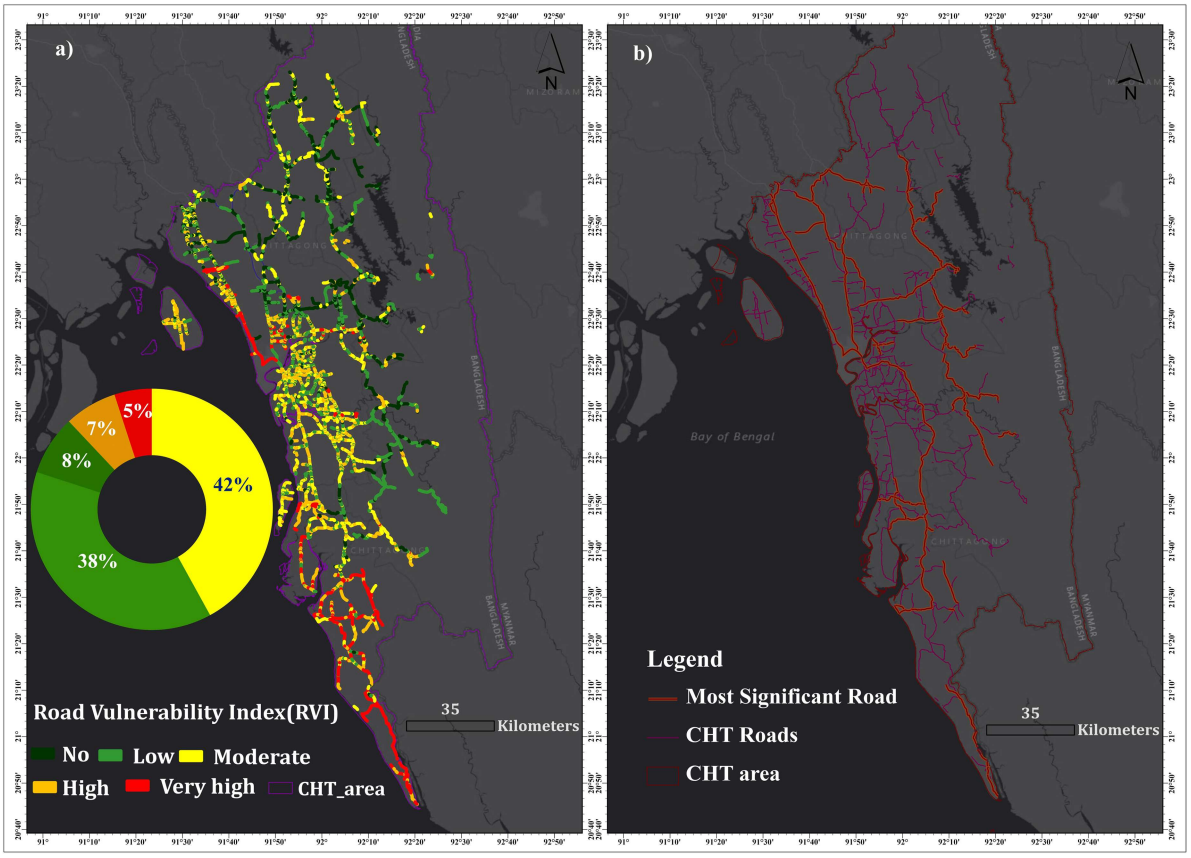


Figure 13. Road vulnerability index for Chittagong Hill Tracks.

region as isolated and mountainous as the CHT, road networks serve as vital conduits for mobility, trade, and emergency response. The model identified over 13% of road segments, including major arteries like the Teknaf Bus Station–Shamlapur Road and Bangladesh Sea Road, as highly or very highly vulnerable. This is particularly concerning given the region's dependence on these corridors for everyday connectivity (Alam and Islam 2023). The RVI introduced in this study therefore provides a practical planning tool, enabling local authorities and engineers to prioritise high-risk segments for reinforcement, realignment, or slope stabilisation. Similar RVI approaches in the Himalayas reduced road closure time by 40% after targeted engineering (Dikshit et al. 2020).

Despite the strengths of this research, several limitations must be acknowledged. Firstly, while precipitation data from sources like CHIRPS and TRMM were interpolated to a 30-metre resolution, these satellite-derived datasets may not fully capture microclimatic variability, particularly in rugged and canopy-dense areas. Secondly, although SHAP enhances the interpretability of the model, it does not offer insights into causal mechanisms, and its effectiveness can be limited when dealing with highly collinear variables, despite efforts to mitigate multicollinearity during preprocessing. Another limitation is the static nature of input data, which means that the model cannot dynamically respond to rapidly changing environmental conditions such as recent deforestation or sudden rainfall bursts. Finally, while the model has shown strong accuracy using historical landslide inventories, underreporting of smaller, undocumented landslides may still introduce spatial bias into the training dataset. Future work should include socio-economic data, as done in Vietnam where adding poverty index improved vulnerability mapping by 25% (Nguyen 2019).

To address these limitations and further enhance the model's utility, future research should consider integrating near-real-time rainfall and vegetation monitoring data through platforms like Google Earth Engine. Additionally, extending the model to incorporate temporal dynamics through the use of spatio-temporal LSTMs or graph-based neural networks could improve its responsiveness to evolving risk

patterns. Another important direction is the inclusion of socio-economic vulnerability indicators, such as population density, poverty levels, or proximity to critical infrastructure like schools and hospitals, to generate a more holistic picture of landslide risk. The development of an interactive, web-based platform to visualise susceptibility and infrastructure vulnerability in real time could further democratise access to this information, enabling more inclusive and informed decision-making. A real-time web platform like the one used in Italy for national landslide warning saved €50 million in one year (Guzzetti et al. 2020). This study demonstrates that a deep learning ensemble model, when combined with rigorous data preprocessing and interpretability tools like SHAP, can effectively capture the complex drivers of landslide susceptibility in a topographically and geologically diverse region like the CHT. By contextualising the model within the region's unique hazard profile and extending its application to critical infrastructure analysis through the RVI framework, this research not only advances methodological approaches but also offers tangible benefits for disaster risk reduction. The framework is ready for immediate use by Bangladesh Road Department and can be adapted to other South Asian hill regions facing similar monsoon and deforestation pressures.

4 Conclusion

This study successfully filled the gap in landslide modelling for the Chittagong Hill Tracts by creating a deep learning ensemble that works well with limited data. The methodology combined high-resolution geospatial datasets with rigorous preprocessing, including normalisation, log transformation, and multicollinearity reduction, ensuring that the selected conditioning features were both statistically robust and contextually relevant. Through stratified sampling and extensive validation, the ensemble model integrating CNN, DNN, and LSTM architectures demonstrated superior performance. The ensemble gave F1-score 0.8923 and AUC 0.9558—better than most recent studies in similar monsoon areas. The force and summary plots indicated that annual rainfall, distance to fault lines, and land use/land cover were the most significant contributors to landslide risk, consistent with historical records and regional hazard literature. Importantly, features such as slope, curvature, and NDVI, while traditionally emphasised, were found to play a more variable role, underscoring the need for context-specific modelling rather than generic assumptions. A major contribution of this work lies in the development of the Road Vulnerability Index (RVI), which translated susceptibility assessments into actionable insights for infrastructure planning. The identification of over 700 road segments as highly or very highly vulnerable including key lifelines such as the Teknaf–Shamlapur Road demonstrates the model's relevance for transport risk mitigation in the CHT. The RVI identified more than 700 high-risk road segments, including Teknaf–Shamlapur Road and Bangladesh Sea Road. This gives clear priority list for urgent slope protection and drainage work. This addition moves beyond static hazard mapping and contributes to resilience planning by highlighting where slope stabilisation and drainage improvements are most urgently required. Despite the model's strong performance, limitations remain. The reliance on satellite-derived rainfall data may overlook local rainfall anomalies, and the absence of real-time updates limits its capacity to track dynamic land use changes. Additionally, while the model integrates a broad range of environmental and geotechnical factors, social vulnerability indicators such as population density or proximity to schools and hospitals were not considered and should be included in future extensions. Adding real-time rainfall and population data will make the model even better for early warning. In conclusion, this research contributes a regionally calibrated, technically rigorous, and operationally useful landslide susceptibility framework for southeastern Bangladesh. By combining methodological innovation with interpretability and infrastructure relevance, the study supports proactive disaster risk reduction and infrastructure resilience in one of the country's most hazard-prone regions. The framework is ready to use by Bangladesh Road Department and can be copied to other hill areas in South Asia facing the same problems.

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Author contributions

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The author declares no competing interests.

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Data availability statement

The data used in this study, including landslide inventory, geospatial datasets, and model outputs, are available from the corresponding author upon reasonable request. Some datasets were derived from publicly available sources such as USGS, TRMM, OpenStreetMap, and Google Earth Engine.

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